MASTERSIZER 2000

THE UNIVERSITY OF WESTERN AUSTRALIA

Sample Name: C_1_2_7 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 4:05:08 PM

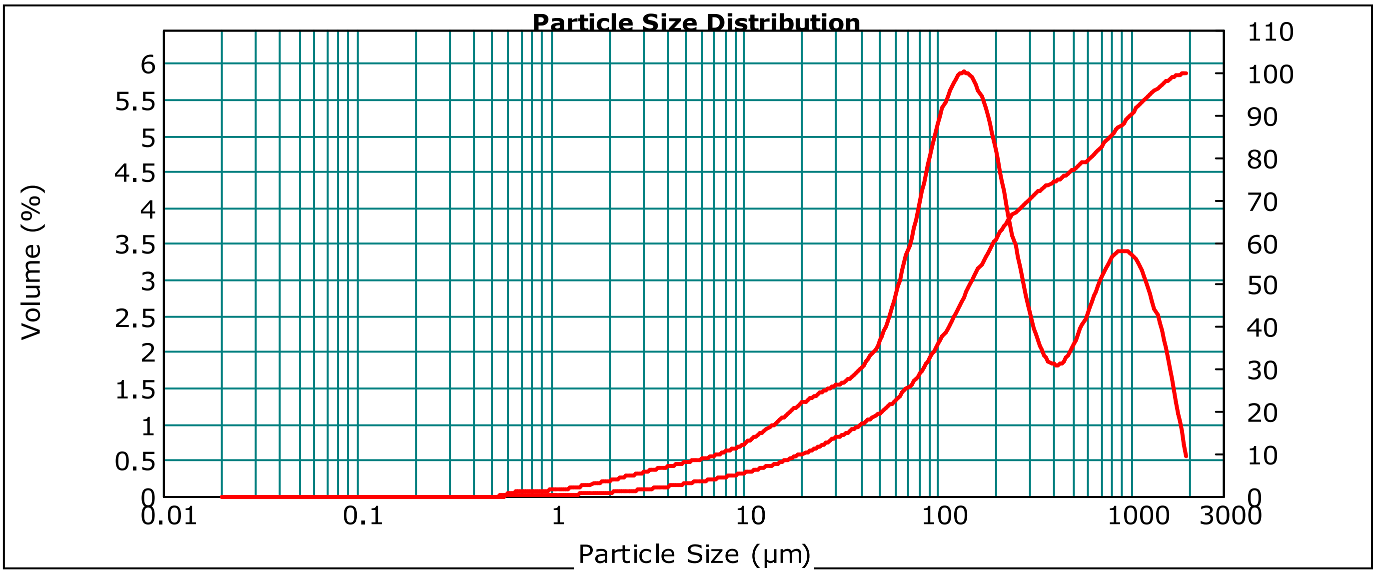
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL alkaline calgon

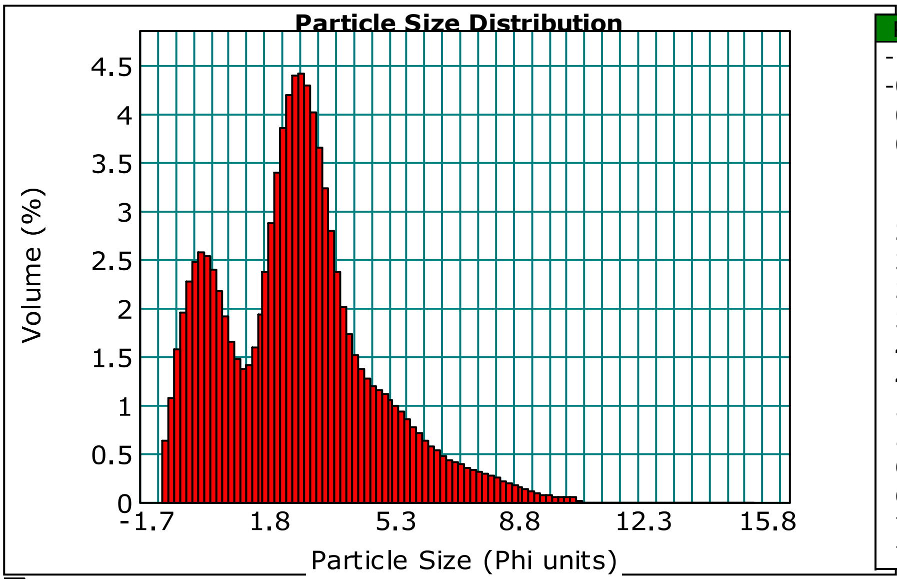
Vol. Weighted Mean:
D[4,3] = 332.99um = 1.59phi
Surface Weighted Mean:
D[3,2] = 35.89um = 4.8phi
Span :
6.580
Uniformity:
1.77
Specific Surface Area*:
0.167 m²/g

Percentiles:
d(0.05) = 9.09um = 6.78phi
d(0.10) = 20.21um = 5.63phi
d(0.16) = 37.58um = 4.73phi
d(0.50) = 149.07um = 2.75phi
d(0.84) = 761.62um = .39phi
d(0.90) = 1001.02um = 0phi
d(0.95) = 1278.01um =-0.35phi

Distribution Properties:
Kurtosis = 2.16
Skewness = 1.71
Standard Deviation = 413.02 um = 1.28phi
Inclusive SD (Sorting Coeff.) = 2.17 phi (Very Poorly
Inclusive Kurtosis = 1.1 phi (Mesokurtic)
Inclusive Mean = 2.62 phi (Fine Sand)
Inclusive Skewness = .02 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (ρ=1). For known densities this needs to be recalculated by dividing by actual density.





Phi	Volume In %
-1.0	3.30
-0.5	6.73
0.0	7.49
0.5	5.73
1.0	4.25
1.5	5.98
2.0	10.22
2.5	13.05
3.0	11.90
3.5	8.28
4.0	5.20
4.5	3.81
5.0	3.29
5.5	2.75
6.0	2.09
6.5	1.55
7.0	1.22
7.5	

Phi	Volume In %
7.5	1.00
8.0	0.79
8.5	0.56
9.0	0.37
9.5	0.23
10.0	0.16
10.5	0.05
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	

MASTERSIZER 2000

THE UNIVERSITY OF WESTERN AUSTRALIA

Sample Name: C_1_2_9 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 1:30:24 PM

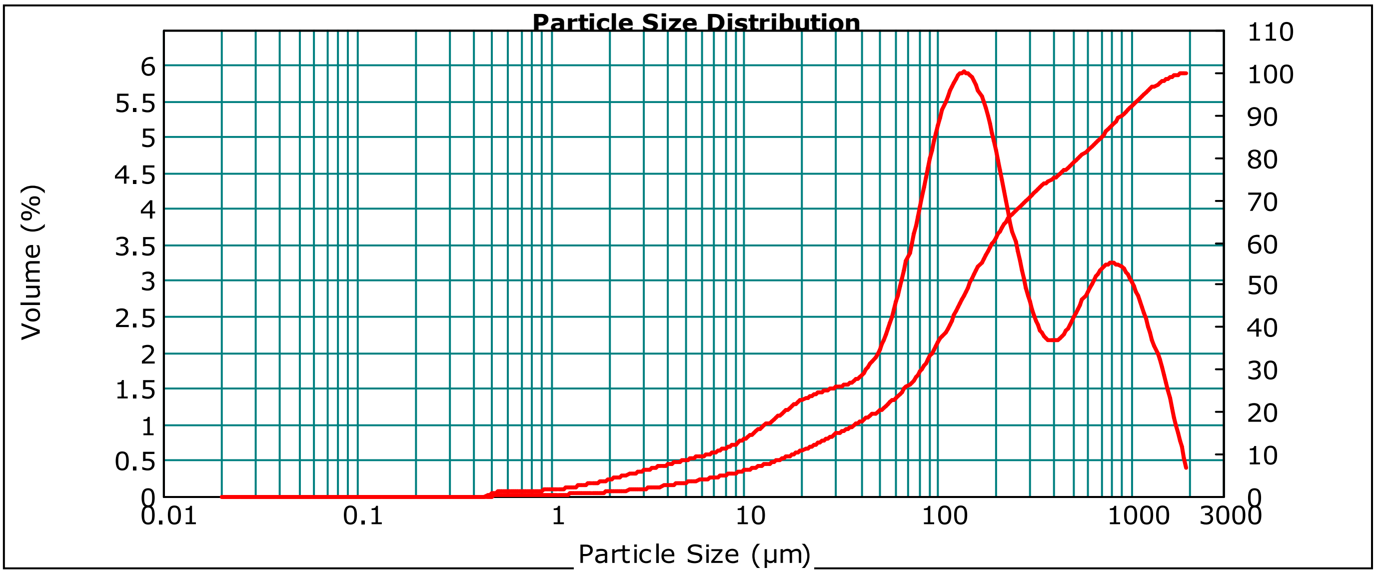
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL alkaline calgon

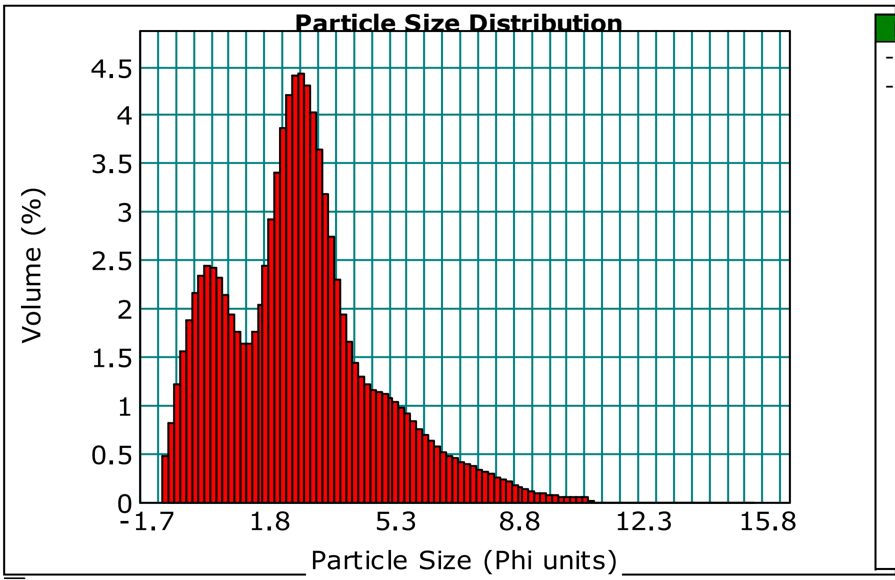
Vol. Weighted Mean:
D[4,3] = 312.5um = 1.68phi
Surface Weighted Mean:
D[3,2] = 32.55um = 4.94phi
Span :
6.042
Uniformity:
1.66
Specific Surface Area*:
0.184 m²/g

Percentiles:
d(0.05) = 8.21um = 6.93phi
d(0.10) = 18.37um = 5.77phi
d(0.16) = 34.82um = 4.84phi
d(0.50) = 147.93um = 2.76phi
d(0.84) = 685.07um = .55phi
d(0.90) = 912.13um = .13phi
d(0.95) = 1189.04um =-0.25phi

Distribution Properties:
Kurtosis = 2.76
Skewness = 1.81
Standard Deviation = 384.94 um = 1.38phi
Inclusive SD (Sorting Coeff.) = 2.16 phi (Very Poorly
Inclusive Kurtosis = 1.15 phi (Leptokurtic)
Inclusive Mean = 2.72 phi (Fine Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (ρ=1). For known densities this needs to be recalculated by dividing by actual density.





Phi	Volume In %
-1.0	2.52
-0.5	5.63
0.0	7.21
0.5	6.39
1.0	5.02
1.5	6.33
2.0	10.29
2.5	13.09
3.0	11.89
3.5	8.11
4.0	4.96
4.5	3.65
5.0	3.30
5.5	2.89
6.0	2.25
6.5	1.68
7.0	1.32
7.5	

Phi	Volume In %
7.5	1.08
8.0	0.85
8.5	0.60
9.0	0.39
9.5	0.24
10.0	0.17
10.5	0.13
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	

MASTERSIZER 2000

THE UNIVERSITY OF WESTERN AUSTRALIA

Sample Name: C_1_2_12 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Monday, 25 August 2025 12:45:12 PM

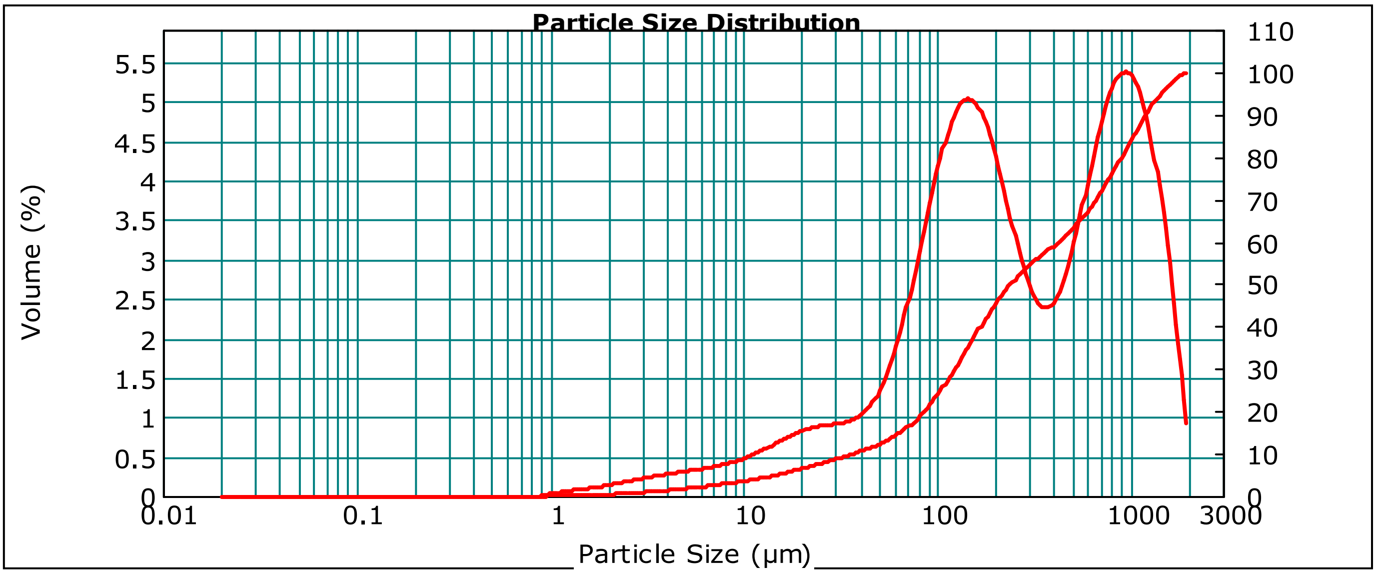
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL alkaline calgon

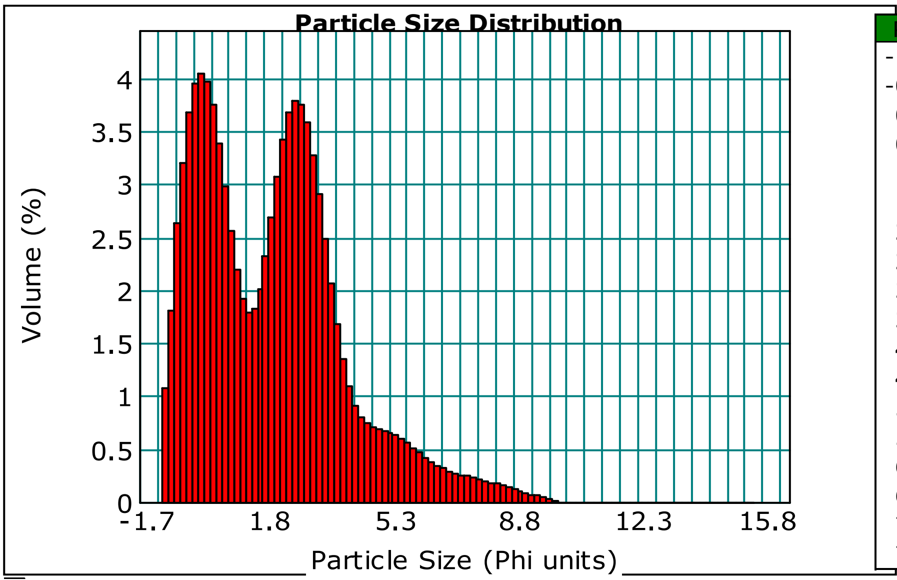
Vol. Weighted Mean:
D[4,3] = 471.12um = 1.09phi
Surface Weighted Mean:
D[3,2] = 57.93um = 4.11phi
Span :
4.867
Uniformity:
1.52
Specific Surface Area*:
0.104 m²/g

Percentiles:
d(0.05) = 15.05um = 6.05phi
d(0.10) = 36.96um = 4.76phi
d(0.16) = 69.58um = 3.85phi
d(0.50) = 240.92um = 2.05phi
d(0.84) = 1011.39um =-0.02phi
d(0.90) = 1209.49um =-0.27phi
d(0.95) = 1443.68um =-0.53phi

Distribution Properties:
Kurtosis = .16
Skewness = 1.08
Standard Deviation = 472.62 um = 1.08phi
Inclusive SD (Sorting Coeff.) = 1.96 phi (Poorly Sorted)
Inclusive Kurtosis = .93 phi (Mesokurtic)
Inclusive Mean = 1.96 phi (Medium Sand)
Inclusive Skewness = .07 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (ρ=1). For known densities this needs to be recalculated by dividing by actual density.





Phi	Volume In %
-1.0	
-0.5	5.52
0.0	10.88
0.5	11.78
1.0	8.90
1.5	5.89
2.0	6.21
2.5	9.28
3.0	11.25
3.5	9.70
4.0	6.10
4.5	3.29
5.0	2.22
5.5	2.02
6.0	1.78
6.5	1.38
7.0	1.02
7.5	0.81

Phi	Volume In %
7.5	
8.0	0.68
8.5	0.54
9.0	0.38
9.5	0.23
10.0	0.12
10.5	0.01
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00

Sample Name: C_1_1_17 - Average
Measured by: WATERCHEM
Particle Name: Unknown Material (RI=1.55 AI=1)
SOP Name: Luka
Refractive Index: 1.550
Absorption Index: 1
Analysis Date: Friday, 22 August 2025 11:43:40 AM

THE UNIVERSITY OF WESTERN AUSTRALIA

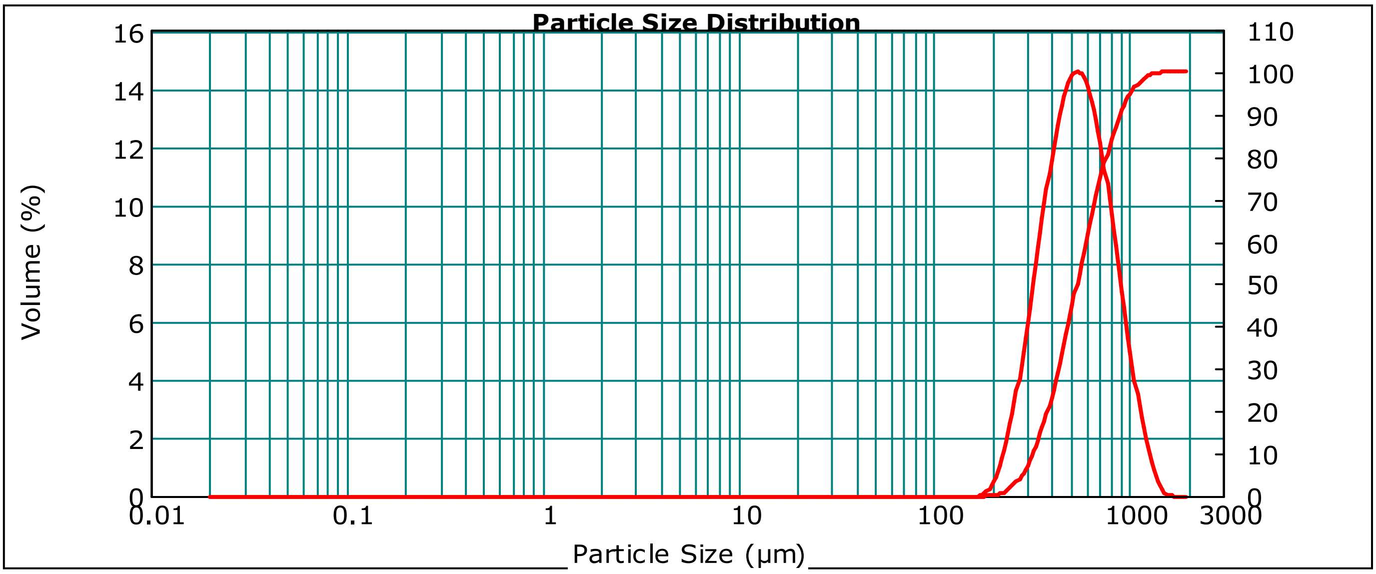
Dispersant: Water
Sonication (seconds): 120
Dispersant Additive: 10mL alkaline calgon

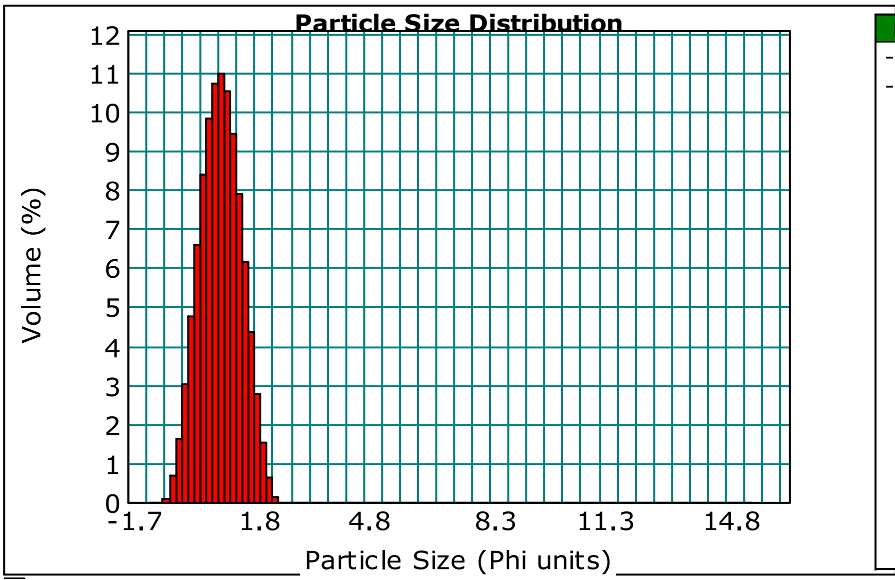
Vol. Weighted Mean:
D[4,3] = 579.3um = .79phi
Surface Weighted Mean:
D[3,2] = 498.46um = 1phi
Span :
1.066
Uniformity:
0.331
Specific Surface Area*:
0.012 m²/g

Percentiles:
d(0.05) = 282.38um = 1.82phi
d(0.10) = 321.42um = 1.64phi
d(0.16) = 358.71um = 1.48phi
d(0.50) = 538.73um = .89phi
d(0.84) = 805.16um = .31phi
d(0.90) = 895.72um = .16phi
d(0.95) = 1014.57um =-0.02phi

Distribution Properties:
Kurtosis = .54
Skewness = .86
Standard Deviation = 227.59 um = 2.14phi
Inclusive SD (Sorting Coeff.) = .57 phi (Moderately Well
Inclusive Kurtosis = .94 phi (Mesokurtic)
Inclusive Mean = .89 phi (Coarse Sand)
Inclusive Skewness = .01 (Near Symmetrical)

* Specific Surface Area is calculated with particle density = 1 (ρ=1). For known densities this needs to be recalculated by dividing by actual density.





Phi	Volume In %
-1.0	
-0.5	0.10
0.0	5.38
0.5	19.91
1.0	31.72
1.5	27.78
2.0	12.99
2.5	2.13
3.0	0.00
3.5	0.00
4.0	0.00
4.5	0.00
5.0	0.00
5.5	0.00
6.0	0.00
6.5	0.00
7.0	0.00
7.5	0.00

Phi	Volume In %
7.5	
8.0	0.00
8.5	0.00
9.0	0.00
9.5	0.00
10.0	0.00
10.5	0.00
11.0	0.00
11.5	0.00
12.0	0.00
12.5	0.00
13.0	0.00
13.5	0.00
14.0	0.00
14.5	0.00
15.0	0.00
15.5	0.00